October 20, 22, 2026

UNIT 2: BONDING

F. Molecular Orbital Theory (8.4)
G. Chemistry of Selected Main Group Elements
H. Acid Strength and Structure (Chapters 14, 15)

**Worksheet on p-Block Elements

Use Molecular Orbital Theory to assess the bonding (bond order and strength) and magnetic properties in second-row homonuclear diatomic molecules and ions.

Utilize different CHEM 103 learning outcomes to predict reactivity and physical properties of main group elements and compounds.

F. Molecular Orbital (MO) Theory (Section 8.4)

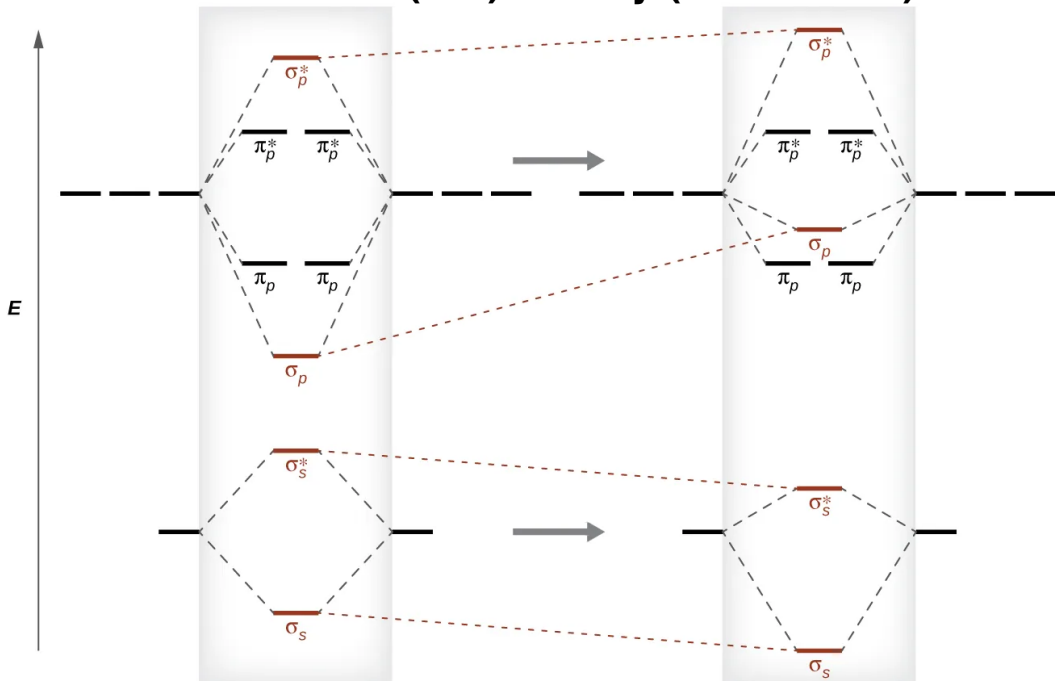
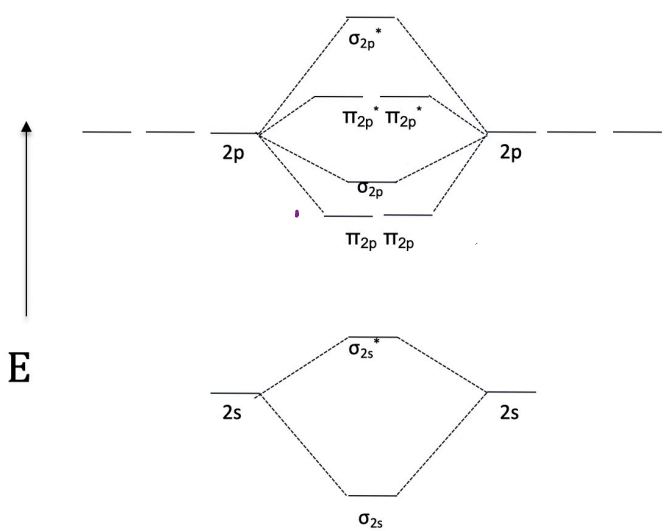


Figure 8.38 Without mixing, the MO pattern occurs as expected, with the σ_p orbital lower in energy than the π_p orbitals. When s-p mixing occurs, the orbitals shift as shown, with the σ_p orbital higher in energy than the π_p orbitals.

MO DIAGRAM for B_2



MO for Diatomic ions: Example N_2^-

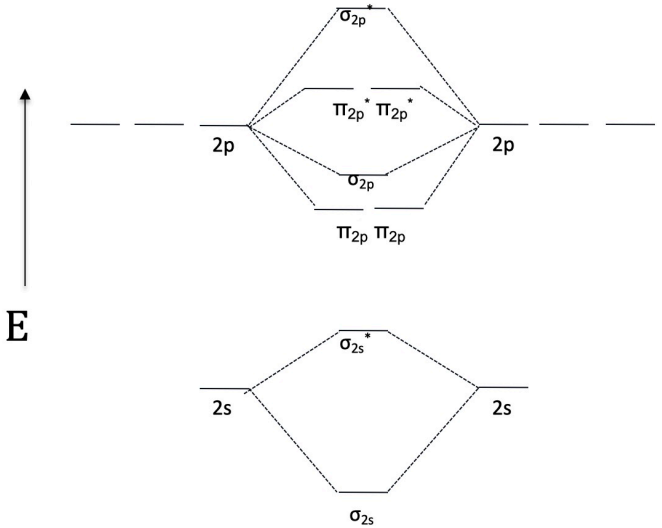
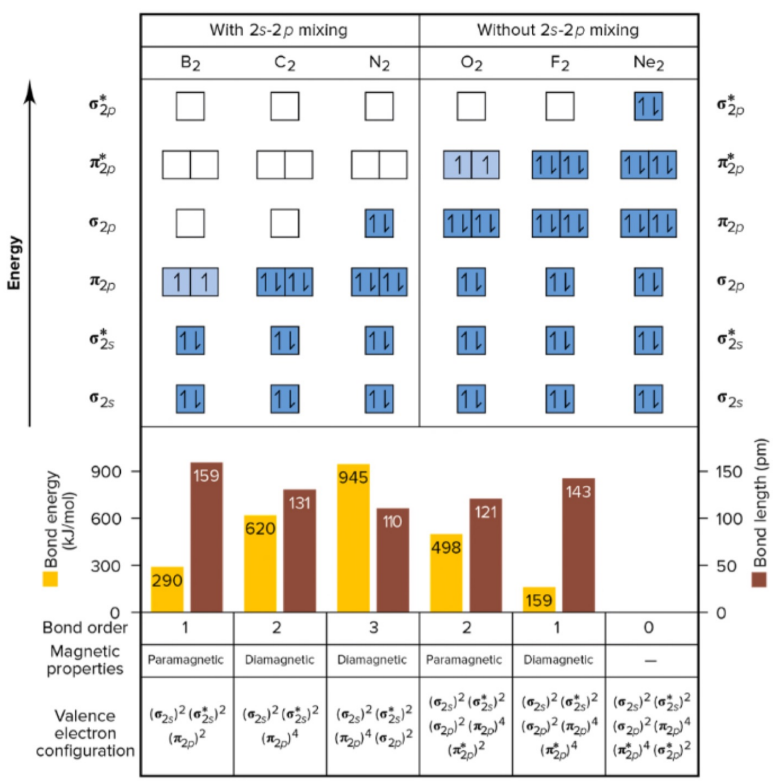


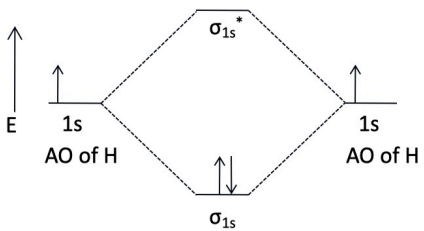
Figure 10.23
MO occupancy and some properties of homonuclear diatomic molecular properties for B_2 through Ne_2.



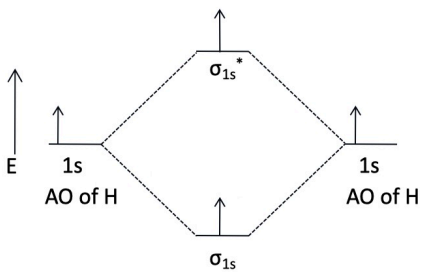
MO Diagrams for Excited State Molecules and Ions

Atoms:	
n = 1	ground state
n = 2	first excited state
n = 3	second excited state
⋮	
⋮	
n = ∞	ionization

Molecules:
Electrons normally **fill the lower energy MOs = Ground state**.
An electron can be promoted to a higher energy MO by absorption of energy = excited state of the molecule.



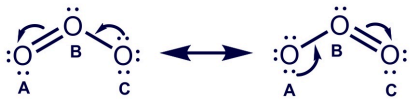
MO of H_2



MO of H_2^*

Valence bond theory treats all bonds as localized, whereas MO theory treats all bonds as completely delocalized. Compare how these theories explain bonding in ozone (O_3) and benzene, C_6H_6.

Valence bond theory:
O_3 can be drawn in 2 ways:



I II

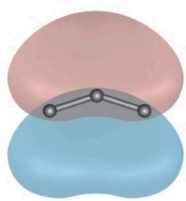
O_3 can be depicted by more than one valid Lewis structure and is best represented as a ***resonance hybrid***.

Resonance forms ***are not real bonding depictions***.

O_3 does ***not*** change back and forth between its two resonance forms.

The ***real structure*** of the resonance hybrid for O_3 is an ***average*** of its contributing resonance forms.

Molecular Orbital Diagram of lowest energy π-bonding MO in ozone: delocalized π bonding electron pair over entire molecule.



Ozone, O_3

Representations of benzene.

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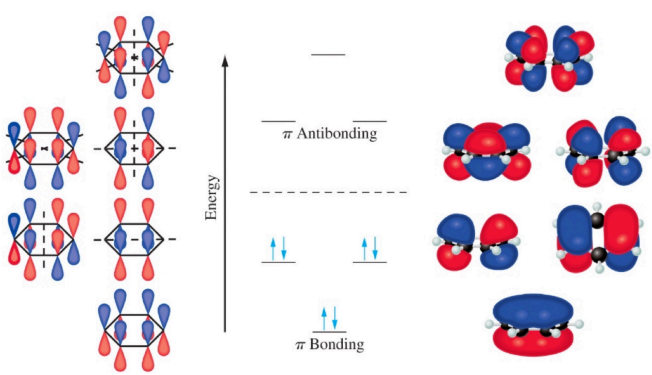
C_6H_6

Resonance forms having alternating single and double bonds.

Resonance hybrid shows the delocalized electrons as either an unbroken or a dashed circle.



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Additional Worked Example on MO Diagram

G. Chemistry of Selected Main Group Elements

1. 2nd-period elements

- **2nd row elements often exhibit distinct properties.**
  - **2^{nd} row elements have high electronegativity**
  - **2^{nd} row elements have small size (n = 2)**
  - **Maximum of four covalent bonds (no d orbitals for bonding)**

- **More common occurrence of *multiple bonding (good π overlap of 2p orbitals)***
  (compare CO_2 vs. SiO_2)

Draw Lewis Structure

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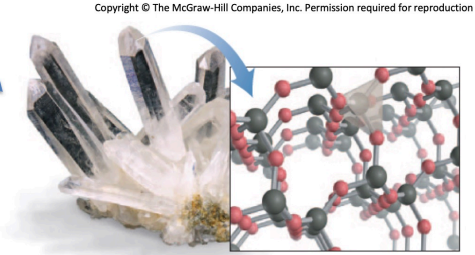
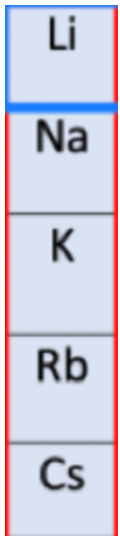


Chains of SiO_4 tetrahedra that share oxygens for empirical formula of SiO_2

Question: Why it is more common to see **multiple bonding** in 2nd row elements than in 3rd row and beyond?

Match the following elements with the type of product (oxide, peroxide, superoxide) they form when reacting with oxygen.



oxide:

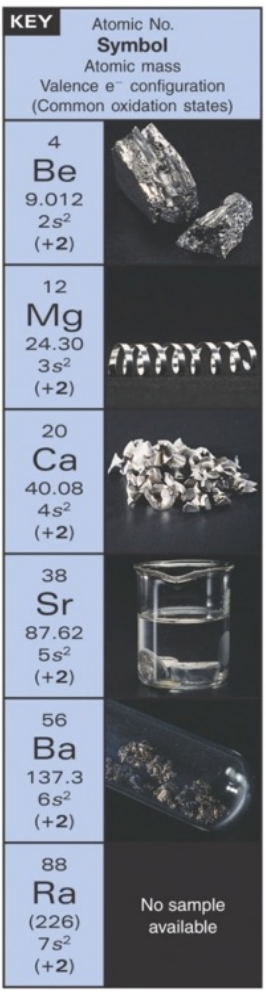
peroxide:

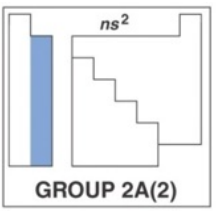
superoxide:

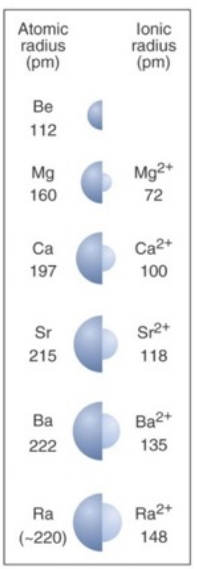
Question: All the halogens can form only one oxidation state (-1). True or False? Explain.

Question: Why are expanded octets and corresponding hybridization sp3d and sp3d2 **not observed** in the 2nd row elements?

2. Group 2: The Alkaline Earth Metals (18.1, 18.2)







- Group 2 elements have higher ionization energies than Group 1 elements → WHY?
- Group 2 elements are **harder and higher-melting** than Group 1 elements.
- Group 2 elements **somewhat less reactive** than Group 1

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Sources of Alkaline Earth Metals

Occurrence: Occur naturally as ionic compounds of M^{2+}.

beryl

magnesite

limestone

celestine

- Natural **carbonates** are major structural materials and the commercial sources for most group 2 compounds. (e.g., limestone, marble)

- Calcium carbonate is heated to obtain calcium oxide (lime). This important industrial compound has essential roles in *steelmaking, water treatment, and smokestack scrubbing, and it is used to make glass, whiten paper, and neutralize acidic soil*.

Preparation. The pure elements can be obtained by ***reduction*** of molten salts.

Summary of Reactions for Group 1 and Group 2 Metals

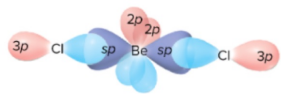
	Group 1	Group 2
Halogens		
Hydrogen		
Water		
Oxygen		
Oxide Behaviour		

Be is unique:

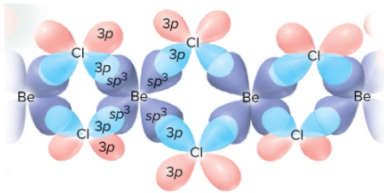
- The element is unreactive in air and water because it forms an oxide coating on the surface.

- BeO is an amphoteric oxide.

- Be compounds have **strong covalent character** and conduct electricity poorly in the molten state.



A



B

Figure 13.3 **Overcoming electron deficiency in beryllium chloride.**

Beryllium chloride ($BeCl_2$): at temperatures greater than 900°C, it consists of linear molecules in which two *sp* hybrid orbitals hold four electrons around the central Be (**Figure 13.3A**). As it cools, the molecules bond together, solidifying in long chains, with each Cl bridging two Be atoms. Each Be is sp^3 hybridized to finally attain an octet (**Figure 13.3B**).

Uses of Alkaline Earth Metals

Be: compounds are toxic, but Be can be used in alloys (Cu-Be, Ni-Be form high strength alloys, used in tools (wrenches, screwdrivers) and other applications.

Mg: Fireworks, lightweight metal alloys (with Al), antacid

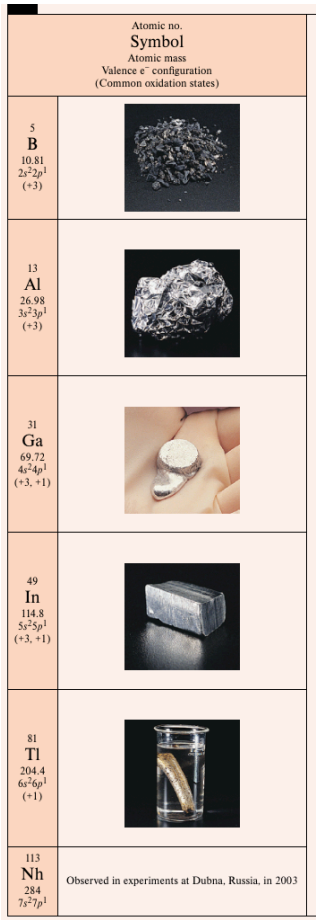
Ca: carbonate salt in limestone, marble and chalk. Used widely in construction, cement and concrete.

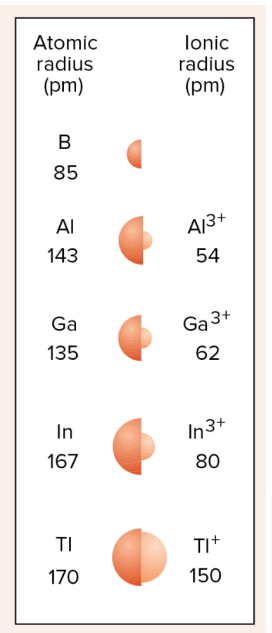
Sr: salts used in fireworks (red color)

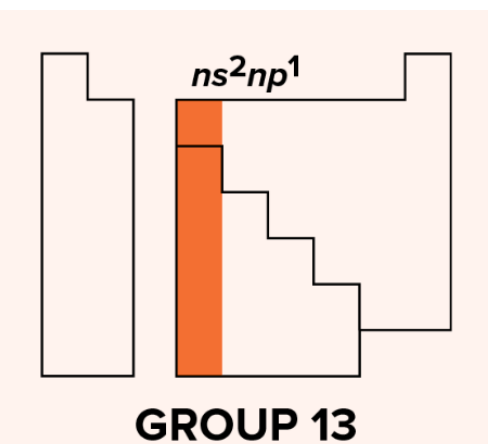
$BaSO_4$ used as Xray contrast ("barium milkshakes")

3. p-Block Elements

Group 3A (13) Examples (boron and aluminum)







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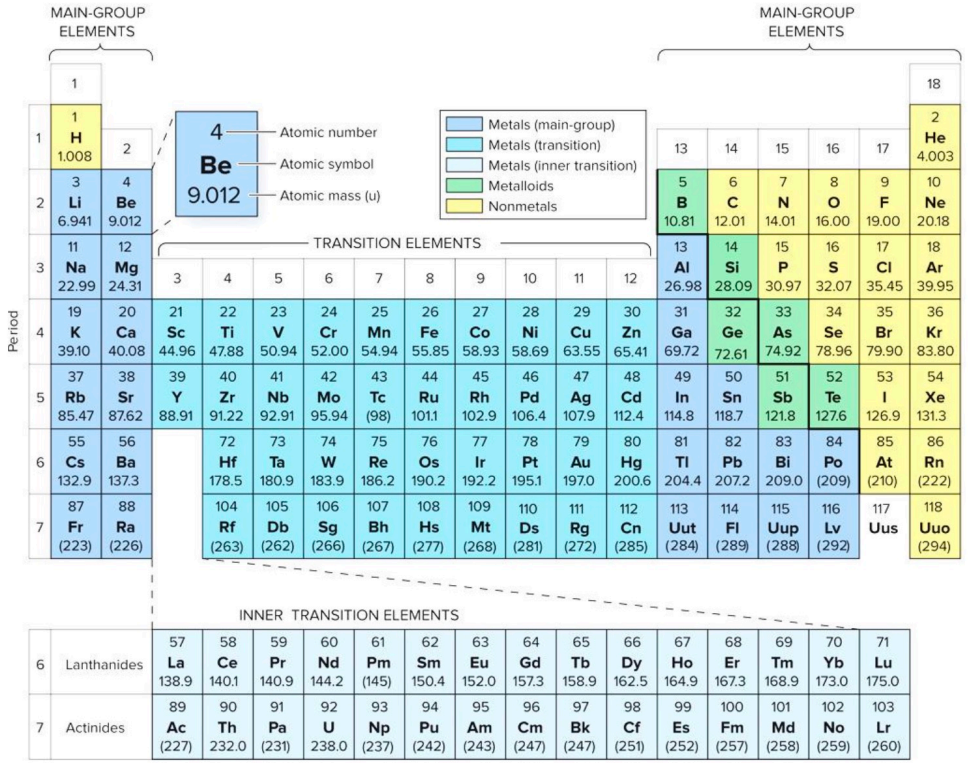
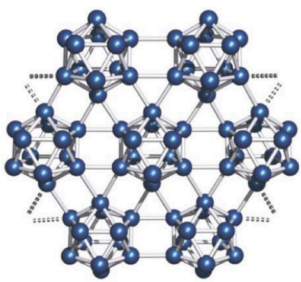

MAIN-GROUP ELEMENTS
MAIN-GROUP ELEMENTS
4
Be
9.012
Atomic number
Atomic symbol
Atomic mass (u)
Metals (main-group)
Metals (transition)
Metals (inner transition)
Metalloids
Nonmetals
TRANSITION ELEMENTS
Period
INNER TRANSITION ELEMENTS
Lanthanides
Actinides

BORON

Metalloid, solid
The structure of its (stable elemental form α-B) contains a **network of covalently bonded** atoms arranged in icosahedral B_{12} units.

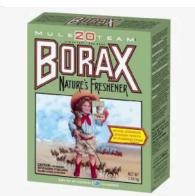
Properties and occurrence. Elemental boron is black, hard, and high melting (mp >2000 ºC). It is a metalloid with a covalent network structure. It occurs naturally in the mineral borax, $Na_2[B_4O_5(OH)_4]\cdot 8H_2O$.

boron crystals



structure of α-boron

borax



20 Mule Team Borax…

Preparation. Borax is converted to the oxide, B_2O_3, and eventually reduced to elemental B with magnesium.

Reactivity

A characteristic feature of many boron-containing compounds is **electron deficiency**, (e.g., BF_3). Consider the **hydride of boron**; with boron having an oxidation state of +3, you might expect its formula to be "BH_3." In fact, the true molecular formula is ***B_2H_6, with two terminal and two bridging H atoms.***

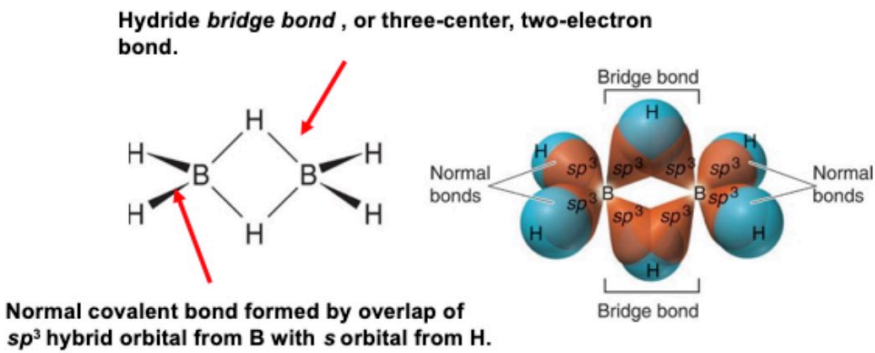
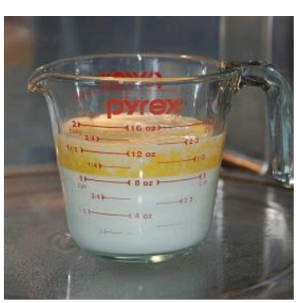
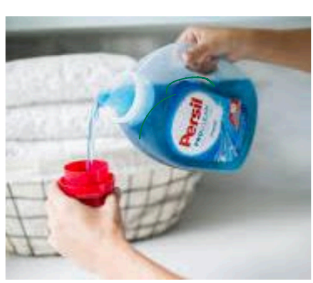


Figure 13.7 The two types of covalent bonding in diborane.

Uses. Some common applications of boron compounds:



Pyrex glass

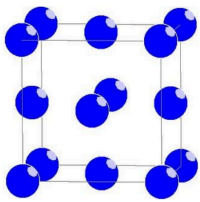


colour-safe bleach

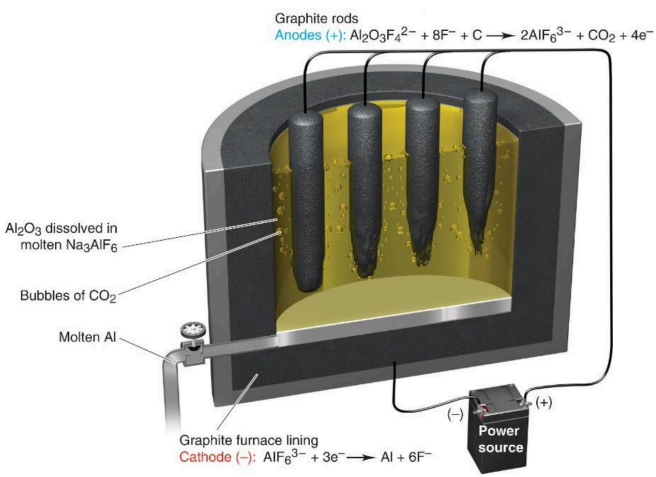
insecticide

ALUMINUM

Properties and occurrence. Aluminum is a typical metal (mp 660 °C), adopting an extended structure. It occurs naturally in the mineral bauxite, a hydrated form of Al_2O_3 (mp 2000 °C).



Preparation. How can Al be extracted from Al_2O_3?

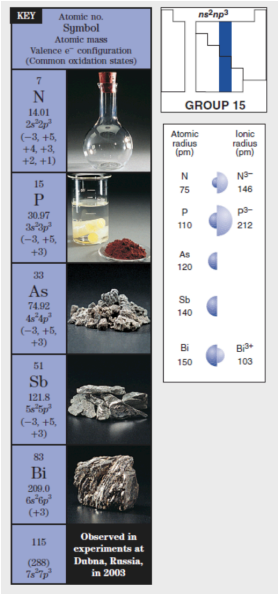


Uses. The light weight and strength of Al make it a desirable component of many structural alloys.

CF-188 Hornet

Reactivity. Aluminum shows representative behaviour in this group, with oxidation state of +3 in its compounds.

GROUP 15 ELEMENTS (PNICTOGENS)



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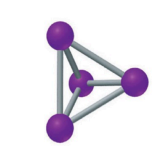
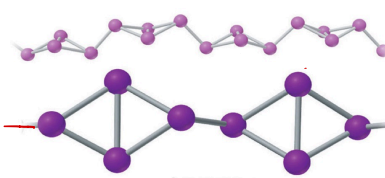
- Nitrogen exists as a triply-bonded diatomic molecule, whereas phosphorus adopts several allotropes containing only single bonds.

$N_2(g)$	$P_4(s)$ (white)	P(s) (red)
colourless gas	white, waxy, toxic, phosphorescent solid	red, nontoxic solid
bp –196 °C	mp 44 °C	mp 600 °C
very inert	highly reactive	less reactive

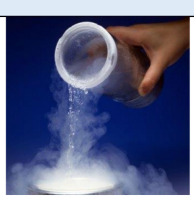


White phosphorous (P_4)



Red phosphorous

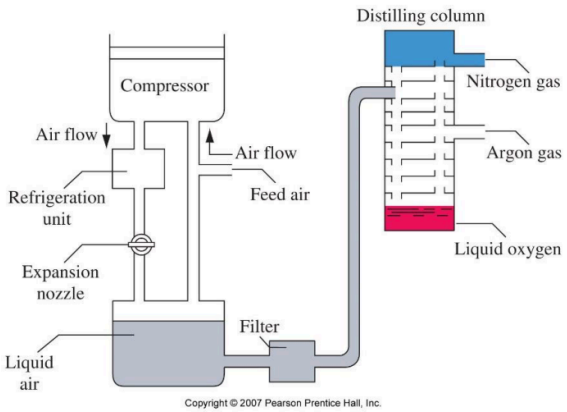
$N_2(g)$	$P_4(s)$ (white)	P(s) (red)



Occurrence and preparation

Nitrogen is abundant in the atmosphere (~80 vol %) and can be isolated by fractional distillation of liquid air.

Video: https://youtu.be/6YzHCSVrUVc



Liquid nitrogen has many uses (as a cryogen, and as an inert atmosphere).

The major naturally occurring source is "phosphate rock," whose composition can be simplified as $Ca_3(PO_4)_2$. How can elemental P be extracted from $Ca_3(PO_4)_2$?

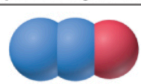
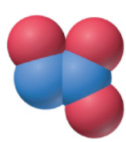
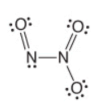
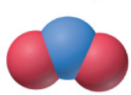
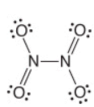
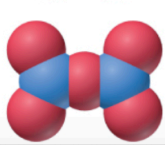
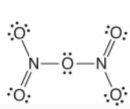
Compounds of nitrogen and phosphorus

(1) Hydrides

$NH_3(g)$	$PH_3(g)$
colourless gas	colourless toxic gas
weak base	weaker base than NH_3
prepared by Haber process	
used as fertilizers, explosives	

(2) Oxides

TABLE 13.3 Structures and Properties of the Nitrogen Oxides

Formula	Name	Space-Filling Model	Lewis Structure	Oxidation State of N	$\Delta_f H$ (kJ/mol) at 298 K	Comment
N_2O	Dinitrogen monoxide (dinitrogen oxide, nitrous oxide)			+1 (0, +2)	82.0	Colourless gas; used as dental anaesthetic ("laughing gas") and aerosol propellant
NO	Nitrogen monoxide (nitrogen oxide, nitric oxide)			+2	90.3	Colourless paramagnetic gas; biochemical messenger; air pollutant
N_2O_3	Dinitrogen trioxide			+3 (+2, +4)	83.7	Reddish-brown gas (reversibly dissociates to NO and NO_2)
NO_2	Nitrogen dioxide			+4	33.2	Orange-brown paramagnetic gas formed during HNO_3 manufacture; poisonous air pollutant
N_2O_4	Dinitrogen tetroxide			+4	9.16	Colourless to yellow liquid (reversibly dissociates to NO_2)
N_2O_5	Dinitrogen pentoxide			+5	11.3	Colourless volatile solid consisting of NO_2^+ and NO_3^-; gas consists of N_2O_5 molecules

Tetraphosphorus hex(a)oxide (or "phosphorus trioxide"), P_4O_6, is a white waxy toxic solid. It is formed when phosphorus reacts with a limited amount of oxygen: $P_4(s)$ + limited $O_2(g) \rightarrow P_4O_6(s)$. Its molecular structure consists of a tetrahedral arrangement of P atoms with six bridging O atoms.

Tetraphosphorus dec(a)oxide (or "phosphorus pentoxide"), P_4O_{10}, is a white solid. It is formed when phosphorus reacts with an excess amount of oxygen: $P_4(s)$ + excess $O_2(g) \rightarrow P_4O_{10}(s)$. Its molecular structure consists of a tetrahedral arrangement of P atoms with six bridging and four terminal O atoms.

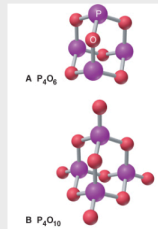


FIGURE 13.19 Important oxides of phosphorus

(3) Oxyacids

Nitrogen and phosphorus oxides are nonmetal oxides, so they dissolve in water to give acidic solutions. (Be able to draw Lewis structures for these oxides and oxyacids.)

$N_2O_3(g) + H_2O(l) \rightarrow$

$N_2O_5(s) + H_2O(l) \rightarrow$

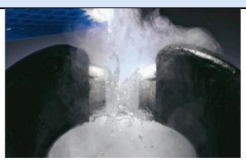
$P_4O_6(s) + H_2O(l) \rightarrow$

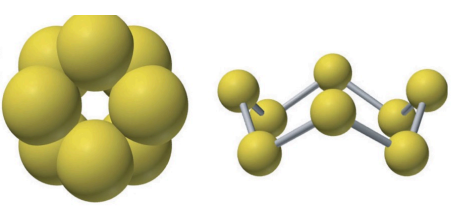
$P_4O_{10}(s) + H_2O(l) \rightarrow$

GROUP 16 ELEMENTS (CHALCOGENS)

Oxygen and sulfur

Properties. Oxygen forms two allotropes, whereas sulfur forms an extraordinary number of allotropes in addition to S_8:

$O_2(g)$, dioxygen	$O_3(g)$, ozone	$S_8(s)$, (rhombic) sulfur
		
colourless, paramagnetic gas	"bluish?" pungent gas	yellow solid
bp –183 °C	bp –112 °C	mp ~ 120 °C
21 vol % in atmosphere	~8 ppm in stratosphere	crown-shaped molecule



At room temperature, the sulfur molecule is a crown-shaped ring of eight atoms. The most stable S allotrope is orthorhombic α-S_8, which consists of cyclo-S_8.

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Occurrence and uses

Oxygen is the most abundant element on the earth's surface. It is readily available in the atmosphere and is isolated by fractional distillation of liquid air. It has many industrial uses (e.g., steel production, chemical manufacture, sewage treatment, oxidizer for rocket fuels).

The earth's crust contains element sulfur and various sulfide minerals (e.g., FeS_2). H_2S and organosulfur compounds are significant impurities in natural gas and crude oil (especially in Alberta), making it difficult to process and refine. What happens when you burn "sour gas" or dirty gasoline?

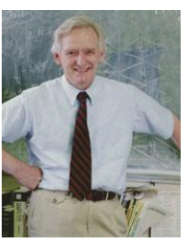
Industrial sources produce the sulfur oxides that lead to acid rain.

Almost all sulfur extracted is destined for preparing sulfuric acid, the world's most important industrial chemical (used as a cheap strong acid in chemical manufacture).

Demo: What happens when you pour sulfuric acid on sugar, $C_{12}H_{22}O_{11}(s)$? https://youtu.be/m_pProDYqTE

GROUP 18 ELEMENTS (NOBLE GASES)

All are monoatomic gases with extremely low mp and bp (e.g., 4 K for He). With completely filled valence shells ($ns^2\ np^6$) and very high IE, these elements are highly unreactive.



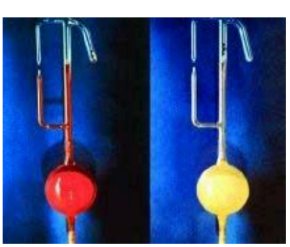
In 1962, Neil Bartlett at UBC was studying the use of PtF_6 as a strong oxidizing agent:

$O_2 + PtF_6 \rightarrow$

He compared the ionization energies of O_2 and Xe:

What if Xe can also be oxidized? From this reasoning, he prepared $XePtF_6$, the first noble-gas compound, as a yellow-orange solid:

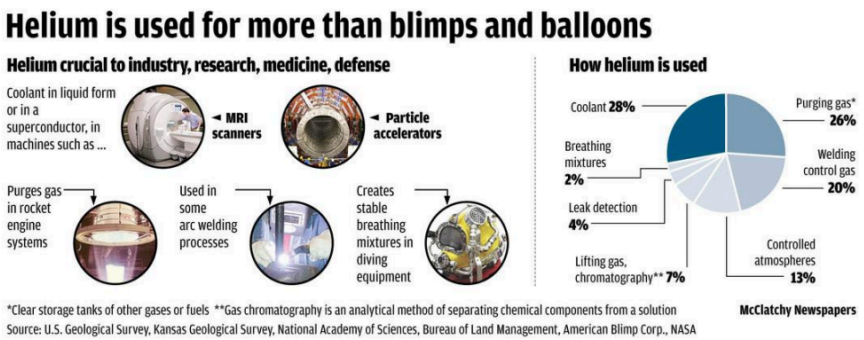
$PtF_6(g)$



$XePtF_6(s)$
Yellow-orange solid

Several fluorides and oxides (XeF_2, XeF_4, XeO_4) are also now known.

Uses. He is an important cryogen, and Ne is used in (what else?) neon signs.



H. Acid Strength and Structure (14.1 - 14.3)

1. Definitions

Arrhenius Definition: acids have H in their formula and form H_3O^+ in aqueous solution. Bases have OH in their formula and form OH^- in aqueous solutions.

Examples: HCl, NaOH

Limitations:

– only applies to aqueous solutions

– definition only includes bases containing OH^- ion

Bronsted-Lowry Definition

acid = proton donor (acid must contain H)

base = proton acceptor (base must contain lone pair to bind H+)

examples: acids = HCl, NH_4^+ and bases = NH_3, NO_2^-

Examples: anything previously identified as an Arrhenius Acid or Base, but with expanded bases (NH_3, F^- =weak base conjugates)

Lewis Definition (15.2)

acid = electron pair acceptor (generally have vacant orbitals to accommodate electrons)

base = electron pair donor (must have electron pairs to donate)

Includes all Bronsted-Lowry acids and bases and includes some new acids because the definition does not require presence of proton.

- The Lewis definition views an acid-base reaction as the ***donation and acceptance of an electron pair*** to form a ***covalent bond***.

- A Lewis base must have a ***lone pair of electrons*** to donate.
  - Any substance that is a Brønsted-Lowry base is also a Lewis base.

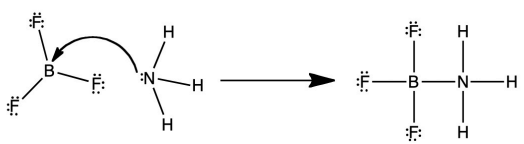
- A Lewis acid must have a ***vacant orbital*** (or be able to rearrange its bonds to form one) to accept a lone pair and form a new bond.
  -Many substances that are **not** Brønsted-Lowry acids are Lewis acids.

- The Lewis definition expands the classes of ***acids***.

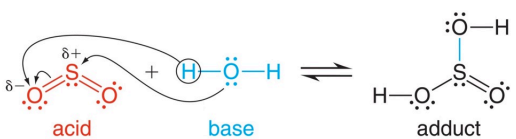
Examples of Lewis acids

(1) Molecules with **electron deficient atoms**

Write the neutralization reaction between $BF_3(g)$ and $NH_3(g)$, to give a Lewis acid-base adduct:

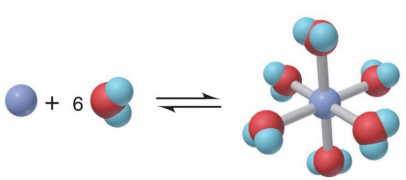


(2) Molecules that contain a **polar multiple bond** often function as Lewis acids



(3) Formation of **coordination complexes**:

$Al^{3+} + 6\ H_2O \rightarrow$



metal cation acts as a Lewis acid

The O atom of an H_2O molecule donates a lone pair to an available orbital on the metal cation.

Metal cations are Lewis acids since they are able to accept electrons. (e.g., Cu^{2+}, $Fe^{2+,}$ $Fe^{3+)}$

An atom, ion, or molecule with an incomplete octet of electrons can act as an Lewis acid (e.g., BF_3, AlF_3).

Molecules where the central atom can have more than 8 valence shell electrons can be electron acceptors, and thus are classified as Lewis acids (e.g., $SiBr_4$, SiF_4).

Molecules that have multiple bonds between two atoms of different electronegativities can act as Lewis acids (e.g., CO_2, SO_2)

An atom, ion, or molecule with a lone-pair of electrons can be a Lewis base. Each of the following anions can "give up" their electrons to an acid, e.g., OH^- , CN^-, CH_3COO^-, NH_3, H_2O, CO

Identify the Lewis acids and bases

$Fe^{3+} + 2H_2O \rightleftharpoons Fe(OH)^{2+} + H_3O^+$

$4CO + Ni \rightleftharpoons Ni(CO)_4$

$H_2SO_4 + NH_3 \rightleftharpoons HSO_4^- + NH_4^+$

$CN^- + H_2O \rightleftharpoons HCN + OH^-$

Bronsted-Lowry Definition

acid = proton donor (acid must contain H)

base = proton acceptor (base must contain lone pair to bind H+)

examples: acids = HCl, NH_4^+ and bases = NH_3, NO_2^-

In an acid-base reaction, a proton is transferred from the acid to the base

$$HA + B \rightleftarrows BH^+ + A^-$$

Write the reaction (in the gas phase) between HCl(g) and NH_3(g):

More typical examples involve reaction with the solvent:

$NH_3(g) + H_2O(l) \rightarrow$

$NH_4^+(aq) + H_2O(l) \rightarrow$

H_3O^+ is called the hydronium ion (is equivalent to H^+).

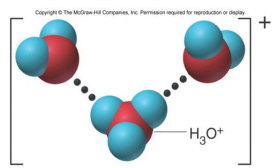
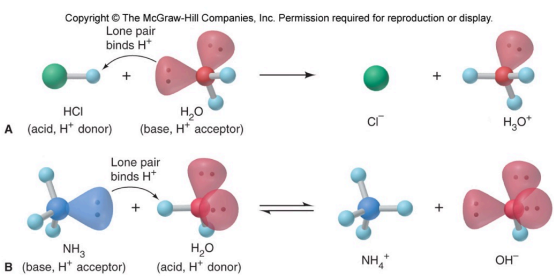


Figure 16.1



Note that water can act as either an acid or a base in a Brønsted-Lowry acid-base reaction.

Figure 16.2

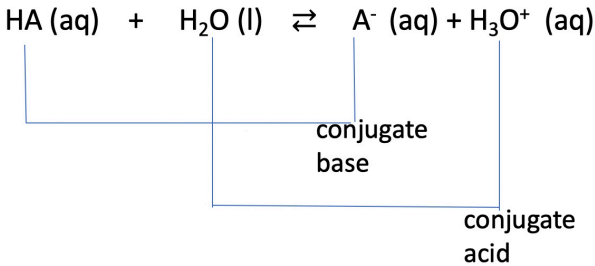
Example: Identify the acid and base, and indicate which definitions apply.

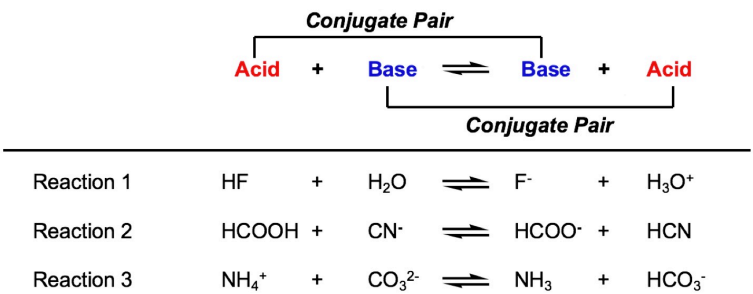
(a) $CaO(s) + SO_2(g) \rightarrow CaSO_3(s)$

(b) $OCl^-(aq) + H_2O(l) \rightleftharpoons HOCl(aq) + OH^-(aq)$

(c) $HCl(aq) + NaOH(aq) \rightarrow NaCl + H_2O$

Conjugate pair = two substances related to each other by gain or loss of a proton.





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Measures of Acid and Base Strength (14.3)

In aqueous solution, we quantify the strength of an acid or a base by relating to the magnitude of the equilibrium constant *K*.

Large K_a = stronger acid, small K_a = weaker acid.

Acid, HA

$HA(aq) + H_2O(l) \rightarrow A^-(aq) + H_3O^+(aq)$

acid ionization constant:

$$K_a = \frac{[A^-][H_3O^+]}{[HA]}$$

Base, B

$B(aq) + H_2O(l) \rightarrow BH^+(aq) + OH^-(aq)$

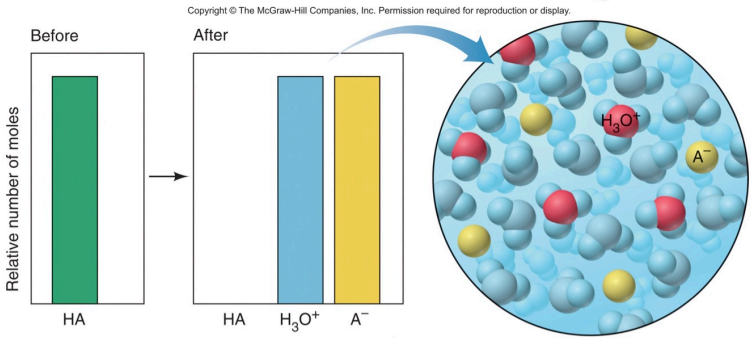
base ionization constant:

$$K_b = \frac{[BH^+][OH^-]}{[B]}$$

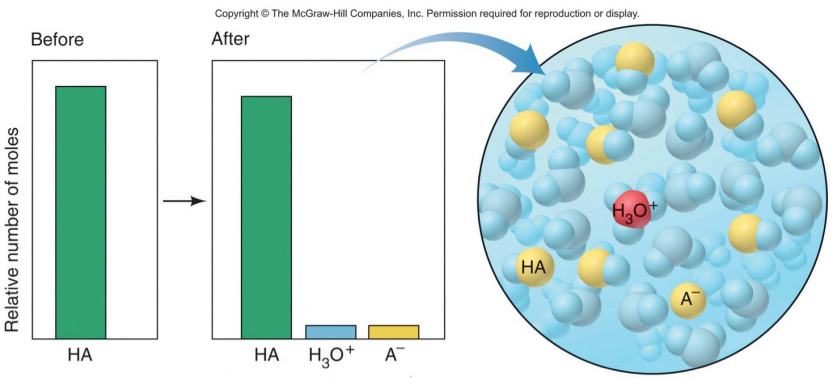
Examples of K_a (aq) for various acids:

HNO_3, nitric acid	very large
HSO_4^-, hydrogen sulfate ion	1.1×10^{-2}
CH_3COOH, acetic acid	1.8×10^{-5}
HCN, hydrocyanic acid	6.2×10^{-10}
H_2O, water	1.0×10^{-14}

Figure 16.4 The extent of dissociation for strong acids.



The extent of dissociation for weak acids.



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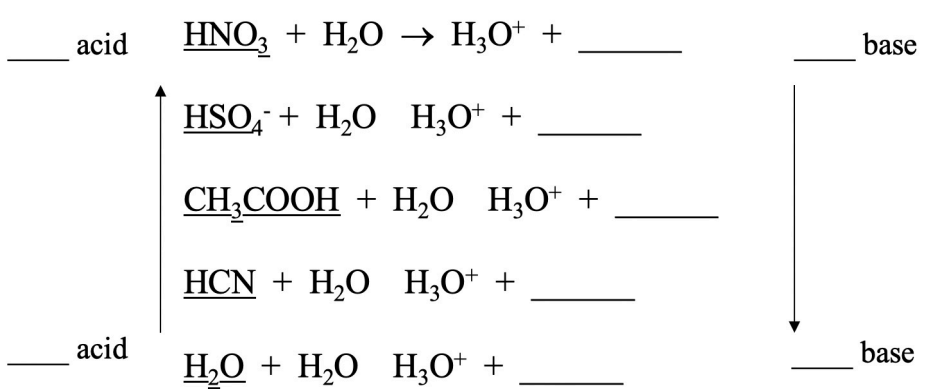
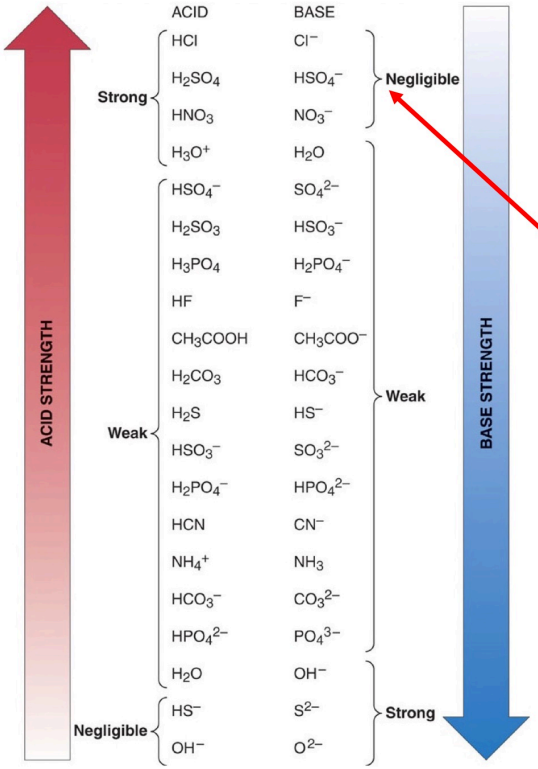
Since acids and bases come in conjugate pairs, we can also deduce the relative strengths of the conjugate bases:





- **Memorize the typical strong acids (see supplementary handout on eClass)** (learn to recognize).
- **The stronger the acid, the weaker the conjugate base.**
- **Conjugates to strong acids are so weak that their strength is negligible.**

Figure 16.3

Strengths of conjugate acid-base pairs.

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Strong Acids and Bases (MEMORIZE)

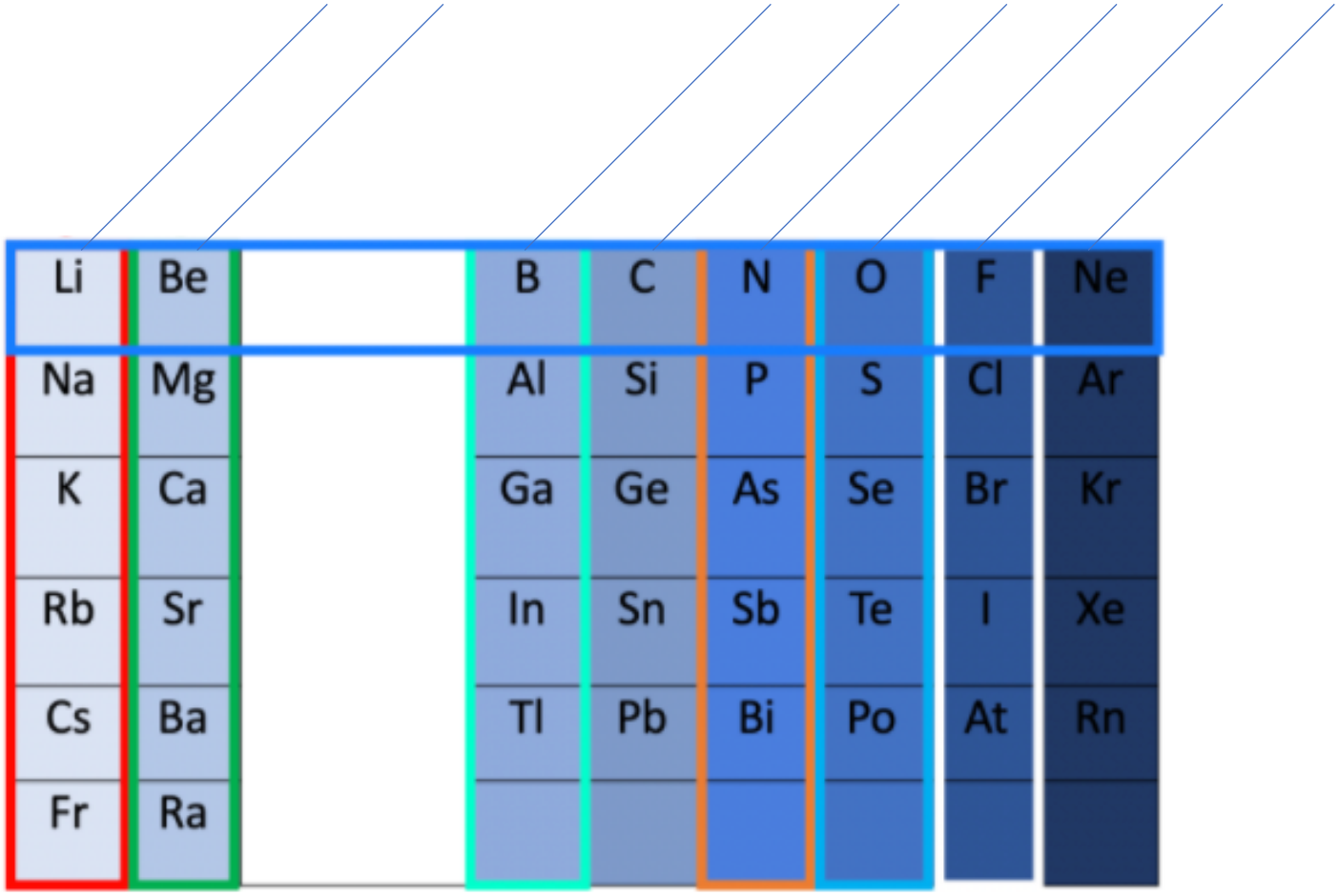
Strong Acids		Strong Bases
HCl	hydrochloric acid	LiOH
HBr	hydrobromic acid	NaOH
HI	hydroiodic acid	KOH
$HClO_4$	perchloric acid	RbOH
HNO_3	nitric acid	CsOH
H_2SO_4	sulfuric acid	$Mg(OH)_2$
		$Ca(OH)_2$
		$Sr(OH)_2$
		$Ba(OH)_2$

WORKSHEET TO GO WITH LESSON ON P-BLOCK ELEMENTS

KEY CONCEPT:

Elements within the same group exhibit _____ behavior because they have _____.

Label the Groups below (use names and Group identification where appropriate):



Metallic behavior **increases/decreases** (circle one) proceeding down a group, therefore **positive/negative** (circle one) oxidation states are prevalent proceeding down a group.

2nd Period elements behave anomalously (why?)—be able to give reasons after completing this lesson.

Anomalous Behaviour of the 2nd-Period Elements:

List the relevant properties to consider:

_____

_____

_____

Match the following elements with the type of product (oxide, peroxide, superoxide) they form when reacting with oxygen.

Oxide O^{2-} Lewis structure for O^{2-} (sketch)

Peroxide O_2^{2-} Lewis structure for O_2^{2-} (sketch)

Superoxide O_2^- Lewis structure O_2^- (sketch)

Question: All the halogens can form only one oxidation state (-1). True or False? Explain.

Question: Explain why expanded octets and corresponding hybridization sp^3d and sp^3d^2 are not observed in the 2nd row elements.

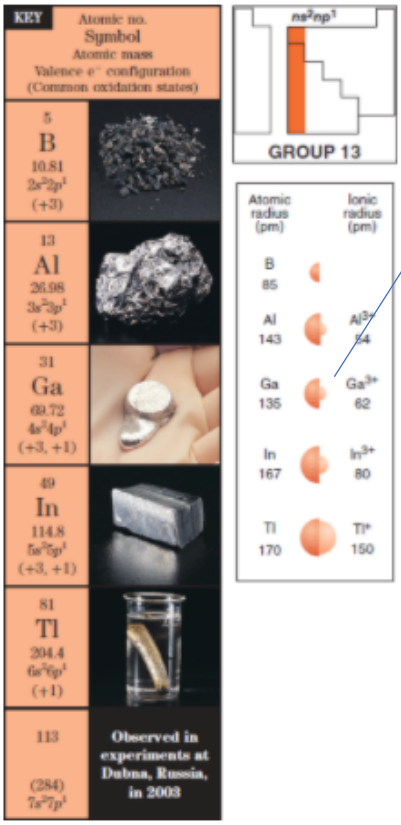
Question: Explain why multiple bonding is observed in 2nd row elements but not in 3rd row and beyond.

Match the **Group 13** elements with the proper label (metal, nonmetal, metalloid).

METAL

NONMETAL

METALLOID



Why does gallium have a smaller radius then aluminum? What is the expected trend? Why is gallium an exception?

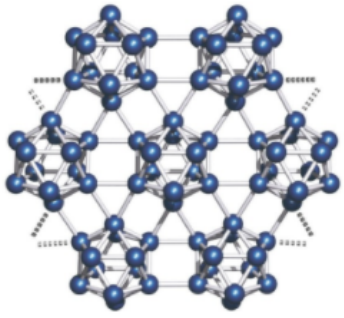
What is the most common oxidation state for B and Al? Why?

In addition to your answer above, what is the additional oxidation state for Ga and In and the most stable oxidation state for Tl. Why is this state observed?

BORON

Metalloid, solid

The structure of its (stable elemental form α-B) contains a **network of covalently bonded** atoms arranged in icosahedral B_{12} units.



(Figure from Housecroft and Sharpe, *Inorganic Chemistry*.)

Note the covalent bonding throughout the structure.
Would you expect boron to have a high or low melting point? Why?

Metalloid
Very hard
Extremely high melting point

Boron is found naturally **as** _____ minerals, such as **borax.** For extraction of boron, these minerals can be converted to B_2O_3 and then **oxidized or reduced** (circle one) **by Al or Mg**.

A characteristic feature of many boron-containing compounds is **electron deficiency**, (e.g., BF_3). Consider the **hydride of boron**; with boron having an oxidation state of +3, you might expect its formula to be "BH_3." In fact, the true molecular formula is ***B_2H_6, with two terminal and two bridging H atoms.***

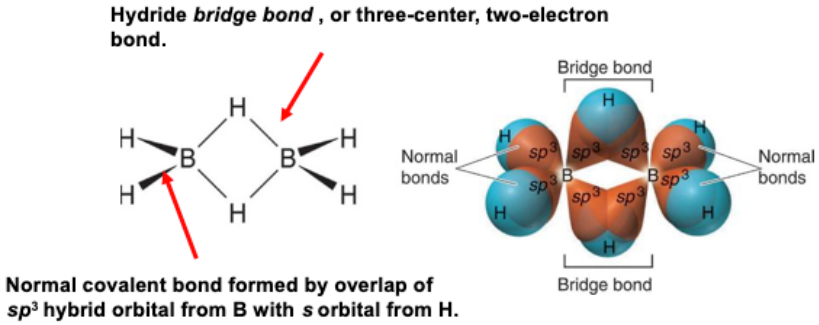
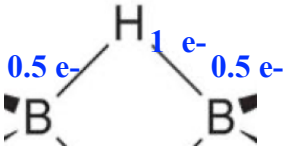


Figure 13.7 The two types of covalent bonding in diborane.

If you attempt to draw Lewis structures, you will quickly discover that it is impossible to satisfy octets around each B atom (the left figure above is *not* a Lewis structure, but simply portrays how the atoms are connected). To account for the bonding in a B-$H_{bridging}$-B group using valence bond theory, **sp^3-hybridized orbitals from each B atom and the 1s orbital from the H atom overlap, and two electrons (0.5 from each B and 1 from H) are distributed over them.** **This three-centre-two-electron bond** illustrates that electrons can be delocalized over more than two atoms, unlike "conventional" two-centre-two-electron bonds (the type that is normally drawn in Lewis structures).



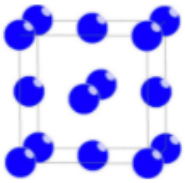
Industrial applications of boron

Boric acid (H_3BO_3 or $B(OH)_3$) is a white solid in its neat form**. List the uses for boric acid:**

Borosilicate glass:

ALUMINUM

Metallic, with metallic crystal structure.



Natural occurrence:

- Most abundant metal in the Earth's crust,
- Occurs as various aluminosilicate minerals (e.g., feldspars, clays). ***Corundum*** **with trace transition metals, exists as the gemstones _____.**

- The major ore of aluminum is a mineral called _____ **a hydrated form of** _____
- Converting $Al_2O_3(s)$ to Al(s) takes place through a **reduction process, carried out by electrolysis**.

  Provide the name of this process: _____

Applications

List **three characteristics** of aluminum alloys that make them particularly valuable in transportation, construction and consumer products: _____ _____ _____

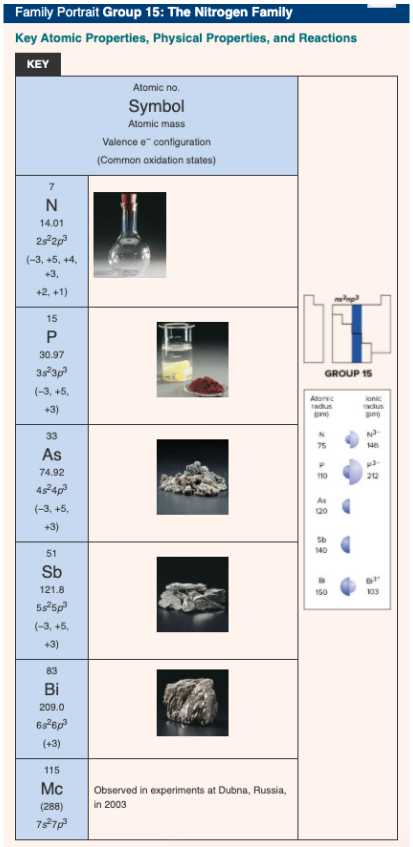
Pnicogens (Group 15)

MATCH

Metal

Nonmetal

Metalloid



Metal + non-metal: to complete octets, the nonmetals nitrogen and phosphorus are likely to form 3- anions in ionic compounds. **Give the formulas and names for the Na and Li ionic compounds formed with N and P.**

Formula **Name**

Formula **Name**

Formula **Name**

Formula **Name**

Nonmetal+non-metal: Commonly, nitrogen and phosphorus combine with many nonmetals to form covalent compounds, and attain oxidation states from **-3 to +5.** On proceeding down the group, the **negative oxidation states become less stable, until +3** is the only common oxidation state for Bi.

NITROGEN $N_2(g)$

- Constitutes about 80% of the atmosphere.
- Isolated via **fractional distillation of liquid air**: separation by boiling off the components of air at different temperatures
- $N_2(g)$ is **highly unreactive**: **describe why N_2 is unreactive**:

- **Do you remember the name of the process that is a primary industrial use of H_2? _____ How is nitrogen used in this process? What is the product? _____**
- Dinitrogen is highly valuable as an **inert blanketing gas** for packaging foods or manufacturing electronic devices, and for performing scientific experiments in glove boxes.
- Low boiling point, liquid nitrogen is used as a **cryogen** (a low-temperature refrigerant) to store temperature-sensitive samples (sperm, eggs).

PHOSPHORUS

- Has many **allotropes** (allotropes = different forms of an element), not all of which are well characterized. Two important ones are white and red phosphorus:

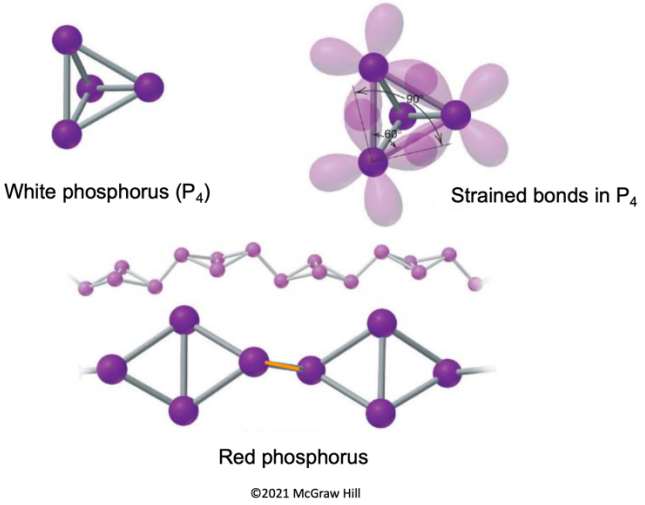


Figure 13.17
Two allotropes of phosphorus.

White phosphorus

- white waxy, highly toxic solid
- low melting point (44 ºC)
- glows faintly
- tetrahedral P_4 clusters with bond angles of 60º
- bonds are "strained" and P_4 is highly reactive.
- ignites on contact with air, and is stored under water
- chemical weapon (*Trigger warning: Search for this information at your own risk*).

Red phosphorus

- red, nontoxic solid
- high melting point (probably around 600 ºC).
- polymeric chains of clusters; bonds are no longer strained. (There is no "molecule" and you can write the formula simply as P(s).)
- less reactive than white phosphorus
- (illegal) manufacture of methamphetamine (watch *Breaking Bad*); also found in the striking surface of safety match boxes; when the match is struck, the red phosphorus transforms to a small amount of white phosphorus, which ignites.)

- Occurs naturally in many phosphate minerals in the earth's crust.
- Major source is phosphate rock, which contains many minerals, including fluoroapatite $Ca_5(PO_4)_3F$ (which also forms part of tooth enamel).
- Extracting the element requires an **oxidation or reduction** (circle one)? _____ **is used as the oxidizing or reducing agent (circle one).**

Hydrides of nitrogen and phosphorus

nitrogen and phosphorus form the expected hydrides with the formula EH_3.

Hydrides

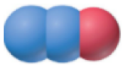
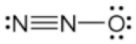
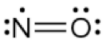
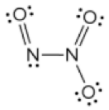
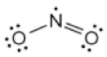
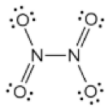
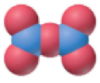
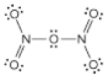
$NH_3(g)$, ammonia	$PH_3(g)$, phosphine
colourless gas	extremely toxic gas
prepared by Haber process	weaker base than NH_3
used as fertilizer and base	

Oxides of nitrogen and phosphorus

Nitrogen

exhibits **positive oxidation states in all its oxides,** as tabulated below. ***Know how to draw the Lewis structures shown and ask yourself if other resonance structures are possible.***

TABLE 13.3 Structures and Properties of the Nitrogen Oxides

Formula	Name	Space-Filling Model	Lewis Structure	Oxidation State of N	$\Delta_f H$ (kJ/mol) at 298 K	Comment
N_2O	Dinitrogen monoxide (dinitrogen oxide, nitrous oxide)			+1 (0, +2)	82.0	Colourless gas; used as dental anaesthetic ("laughing gas") and aerosol propellant
NO	Nitrogen monoxide (nitrogen oxide, nitric oxide)			+2	90.3	Colourless paramagnetic gas; biochemical messenger; air pollutant
N_2O_3	Dinitrogen trioxide			+3 (+2, +4)	83.7	Reddish-brown gas (reversibly dissociates to NO and NO_2)
NO_2	Nitrogen dioxide			+4	33.2	Orange-brown paramagnetic gas formed during HNO_3 manufacture; poisonous air pollutant
N_2O_4	Dinitrogen tetroxide			+4	9.16	Colourless to yellow liquid (reversibly dissociates to NO_2)
N_2O_5	Dinitrogen pentoxide			+5	11.3	Colourless volatile solid consisting of NO_2^+ and NO_3^-; gas consists of N_2O_5 molecules

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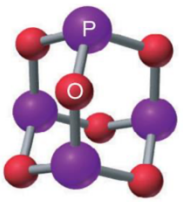
Phosphorus

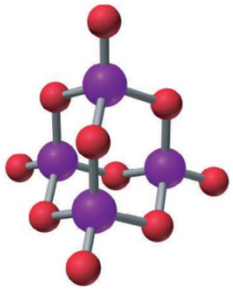
Exhibits **oxidation numbers +3 and +5 in its two common oxides.**

Tetraphosphorus hex(a)oxide (or "phosphorus trioxide"), P_4O_6, is a white waxy toxic solid.
Tetraphosphorus dec(a)oxide (or "phosphorus pentoxide"), P_4O_{10}, is a white solid.

Figure 13.19 Important oxides of phosphorus.



P_4O_6 has P in its +3 oxidation state.



P_4O_{10} has P in its +5 oxidation state.
This compound is a powerful drying agent.

$N_2O(g)$, dinitrogen monoxide	colourless, sweet-smelling gas dental anesthetic ("laughing gas") propellant for whipped cream
$NO(g)$, nitrogen monoxide	colourless, paramagnetic gas air pollutant, neural transmitter
$NO_2(g)$, nitrogen dioxide	brown, toxic gas; implicated in photochemical smog formed by reaction of $NO + O_2$
P_4O_6, "phosphorus trioxide"	$P_4(s)$ + limited $O_2(g) \rightarrow P_4O_6(s)$
P_4O_{10}, "phosphorus pentoxide"	$P_4(s)$ + excess $O_2(g) \rightarrow P_4O_{10}(s)$

Oxyacids of nitrogen and phosphorus

Because nitrogen and phosphorus oxides are nonmetal oxides, they react with water to form acidic solutions. ***Recall the key principle that the oxidation number remains unchanged in the oxide and its corresponding oxyacid***.

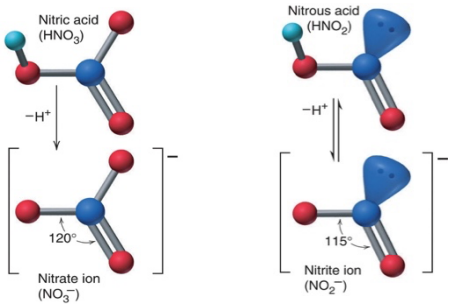
Nitrous acid, $HNO_2(aq)$

- **Strong or weak (circle one) acid**
- **N in the _____ oxidation state.**
- **$N_2O_3(g) + H_2O(l) \rightarrow 2\ HNO_2(aq)$**
- It is a useful reagent in organic synthesis.
- The conjugate base of nitrous acid is the **nitrite ion, NO_2^-**

Nitric acid, $HNO_3(aq)$

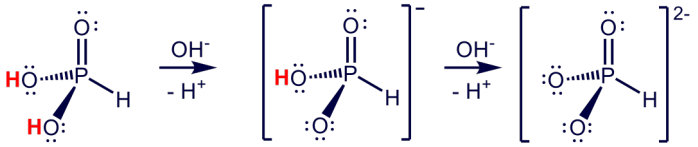
- **Strong or weak (circle one) acid**
- **N in the _____ oxidation state**.
- **$N_2O_5(s) + H_2O(l) \rightarrow 2\ HNO_3(aq)$**
- **Industrial preparation (_____ process)**
- The major industrial application of nitric acid is the acid-base reaction with ammonium nitrate to form **ammonium nitrate**, $HNO_3(aq) + NH_3(aq) \rightarrow NH_4NO_3(s)$, used as a **fertilizer**
- The conjugate base of nitric acid is the **nitrate ion, NO_3^-**.
- **Very strong oxidizing agent.**

Figure 14.17 The structures of nitric and nitrous acids



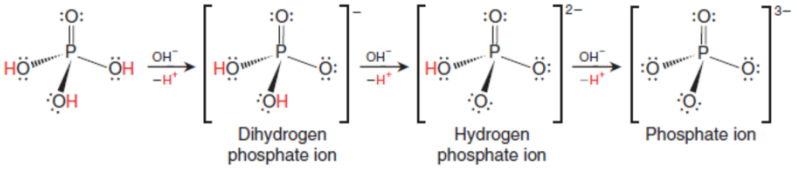
Phosphorous acid, $H_3PO_3(aq)$

- **Strong or weak (circle one) acid**
- **P in the _____ oxidation state**
- **$P_4O_6(s) + 6\ H_2O(l) \rightarrow 4\ H_3PO_3(aq)$,** but this is not the industrial route.
- **Not a triprotic acid, but a diprotic acid: one of the H atoms is directly bonded to the central P atom and is not acidic.**



Phosphoric acid, $H_3PO_4(aq)$,

- **Strong or weak (circle one) acid**
- **P in the _____ oxidation state**
- **$P_4O_{10}(s) + 6\ H_2O(l) \rightarrow 4\ H_3PO_4(aq)$** but this is not the industrial route.



- **It is a triprotic acid, with three acidic H atoms that could potentially dissociate**.
- The main industrial application is as a starting material for fertilizers.

Nitrogen and phosphorus oxides are ***nonmetal oxides***, so they dissolve in water to give acidic solutions. (Be able to draw Lewis structures for these oxides and oxyacids.) Balance the below reactions and draw the Lewis structures for the product acids.

$N_2O_3(g) + H_2O(l) \rightarrow$

$N_2O_5(s) + H_2O(l) \rightarrow$

$P_4O_6(s) + H_2O(l) \rightarrow$

$P_4O_{10}(s) + H_2O(l) \rightarrow$

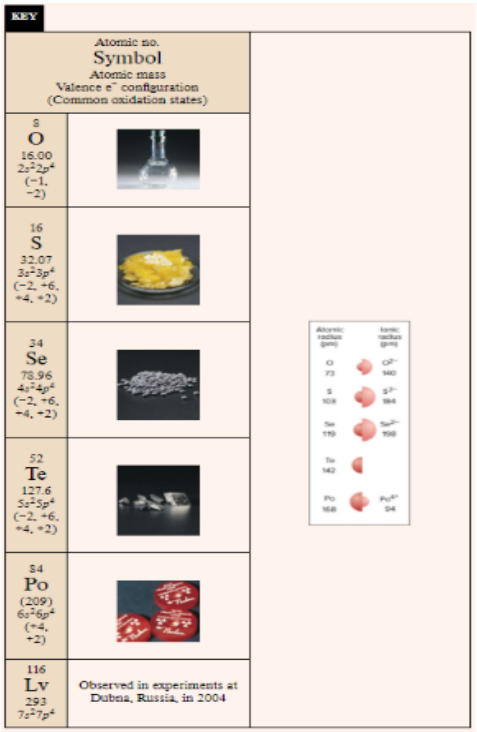
Group-16 elements(chalcogens)

MATCH

Metal

Nonmetal

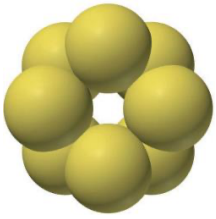
Metalloid



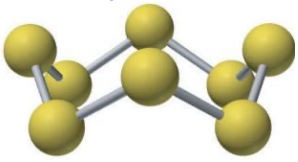
- Electron configuration $ns^2\ np^4$.
- Nonmetals tend to form 2- ions in ionic compounds (e.g., MgO, ZnS).
- In covalent compounds, sulfur exhibits oxidation states from -2 to +6.
- On proceeding down the group, the positive oxidation states become more stable for Te.

$O_2(g)$, dioxygen	$O_3(g)$, ozone	$S_8(s)$, (rhombic) sulfur
• colourless, paramagnetic gas • most abundant element on the Earth's surface. • bp –183 °C • 21 vol % in atmosphere • **Paramagnetic or diamagnetic ? (circle one) Hint use MO diagram** • **Extracted via _____ _____**	• bluish, pungent gas • bp –112 °C • ~8 ppm in stratosphere • Diamagnetic • Absorbs UV in upper atmosphere • Lower atmosphere = pollutant	• yellow solid • **sulfide minerals** are abundant source of sulfur. • mp ~ 120 °C • crown-shaped molecule • sp^3 hybridized S

Figure 13.21 The cyclo-S_8 molecule.



top view



side view

- Oxygen is the most abundant element on the earth's surface. It has many industrial uses (e.g., steel production, chemical manufacture, sewage treatment, oxidizer for rocket fuels).
- Sulfur-containing compounds are contaminants in fossil fuels (coal, oil sands, natural gas) and must be removed to avoid combustion to $SO_2(g)$, which would eventually oxidize to $SO_3(g)$ and combine with moisture to **form acid rain**:

  $S(s) + O_2(g) \rightarrow SO_2(g)$

  $SO_2(g) + ½O_2(g) \rightarrow SO_3(g)$

  $SO_3(g) + H_2O(g) \rightarrow H_2SO_4(aq)$
- Major commercial application of sulfur is in the **production of _____ which is by far the most important chemical substance in industry** where it is used as a **strong acid** in chemical manufacture (especially fertilizers).

Name and draw Lewis structures for the two sulfur oxyacids:

Strong or weak acid?

Strong or weak acid?

- Sulfuric acid is a strong **dehydrating and oxidizing agent**. Watch this video of a standard lecture demonstration showing what happens when sulfuric acid is added to sucrose: https://www.youtube.com/watch?v=m_pProDYqTE